🚚 Supply Chain & Warehouse Optimization System (SCWOS)

Language: ANSI C
Architecture: Structured Enterprise Console System
Storage: Binary File-Based Persistence
Version: 1.0

📌 Executive Summary

The Supply Chain & Warehouse Optimization System (SCWOS) is a structured logistics and inventory management platform developed entirely in ANSI C.

The system integrates core supply chain operations including:

• Secure administrator authentication
• Product inventory management
• Supplier tracking
• Shipment processing
• Cost analytics
• Reorder monitoring
• Timestamp-based audit logging

SCWOS models warehouse and supply chain workflow using deterministic structured programming and binary file-based persistence without reliance on external databases.

🏗️ System Architecture

SCWOS follows a layered logical structure:

• Authentication Layer
• Product Inventory Module
• Supplier Management Module
• Shipment Processing Module
• Cost Analytics Engine
• Audit Logging Layer
• Binary Persistence Layer

Although implemented in a single source file, functional responsibilities remain logically separated.

🔐 Authentication Module

The system ensures secure administrative access.

Features:

• Automatic admin initialization on first execution
• Password hashing using djb2 algorithm
• No plaintext credential storage
• Binary verification against `supply_users.dat`

Default Credentials:

Username: admin
Password: admin123

Hashing Logic:

```
hash = ((hash << 5) + hash) + character
```

📦 Product Inventory Management

Each product record contains:

• Product ID
• Product Name
• Quantity
• Cost Per Unit
• Reorder Level

Functional Capabilities:

• Add product
• View product inventory
• Detect low stock (quantity ≤ reorder level)

When quantity reaches or falls below reorder level, the system displays:

REORDER REQUIRED

Storage File: `products.dat`

🤝 Supplier Management

Each supplier record contains:

• Supplier ID
• Supplier Name
• Reliability Score (0–100)

Functional Capabilities:

• Add supplier

Storage File: `suppliers.dat`

🚚 Shipment Processing Module

Each shipment record contains:

• Product ID
• Shipment Quantity
• Total Cost
• Timestamp

Operational Flow:

• Validate product existence
• Increase inventory quantity
• Compute total shipment cost
• Record shipment with timestamp
• Persist record

Total Cost Formula:

```
Total Cost = Quantity × Cost Per Unit
```

Storage File: `shipments.dat`

📊 Cost Analytics Engine

The analytics module processes recorded shipment data and computes:

• Mean (Average Shipment Cost)
• Variance
• Standard Deviation

Formulas Used:

```
Mean = Σ Shipment Costs / N
Variance = Σ (Cost − Mean)² / N
Standard Deviation = √Variance
```

Purpose:

• Evaluate shipment cost distribution
• Measure procurement variability
• Analyze warehouse financial flow

📝 Audit Logging

All major system activities are recorded in:

`supply_logs.txt`

Logged Events:

• Admin login
• Product addition
• Supplier addition
• Shipment recording
• Analytics execution
• System exit

Each log entry includes timestamp metadata for traceability.

💾 Data Storage Design

Binary files used:

File	Purpose
supply_users.dat	Authentication records
products.dat	Product inventory
suppliers.dat	Supplier records
shipments.dat	Shipment records

File	Purpose
supply_logs.txt	Audit logs

Binary storage ensures:

• Structured record alignment
• Faster read/write operations
• Reduced storage overhead
• Persistent inventory integrity

⚙️ Compilation & Execution

Compile:

```
gcc supply_chain_system.c -o supply -lm
```

Run:

```
./supply
```

🧠 Engineering Highlights

• ANSI C structured programming
• File-based warehouse database simulation
• Real-time inventory updates
• Reorder detection logic
• Shipment cost computation
• Statistical cost analysis
• Deterministic memory usage
• Timestamp-based traceability

🎯 Project Significance

SCWOS demonstrates the capability to design and implement a centralized warehouse and supply chain management system using low-level systems programming.

It reflects:

• Logistics workflow modeling
• Structured data persistence design

• Cost optimization analysis
• Secure authentication handling
• Enterprise-grade inventory logic

🏁 Conclusion

The Supply Chain & Warehouse Optimization System (SCWOS) is a complete logistics enterprise simulation platform implemented in pure ANSI C.

It integrates:

Authentication
Inventory Management
Supplier Tracking
Shipment Processing
Cost Analytics
Audit Logging

All within a structured and deterministic runtime architecture.